



PRINCIPLES OF FLIGHT

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GENERAL REVIEW

BASIC AERODYNAMIC THEORY



The Principle of Continuity

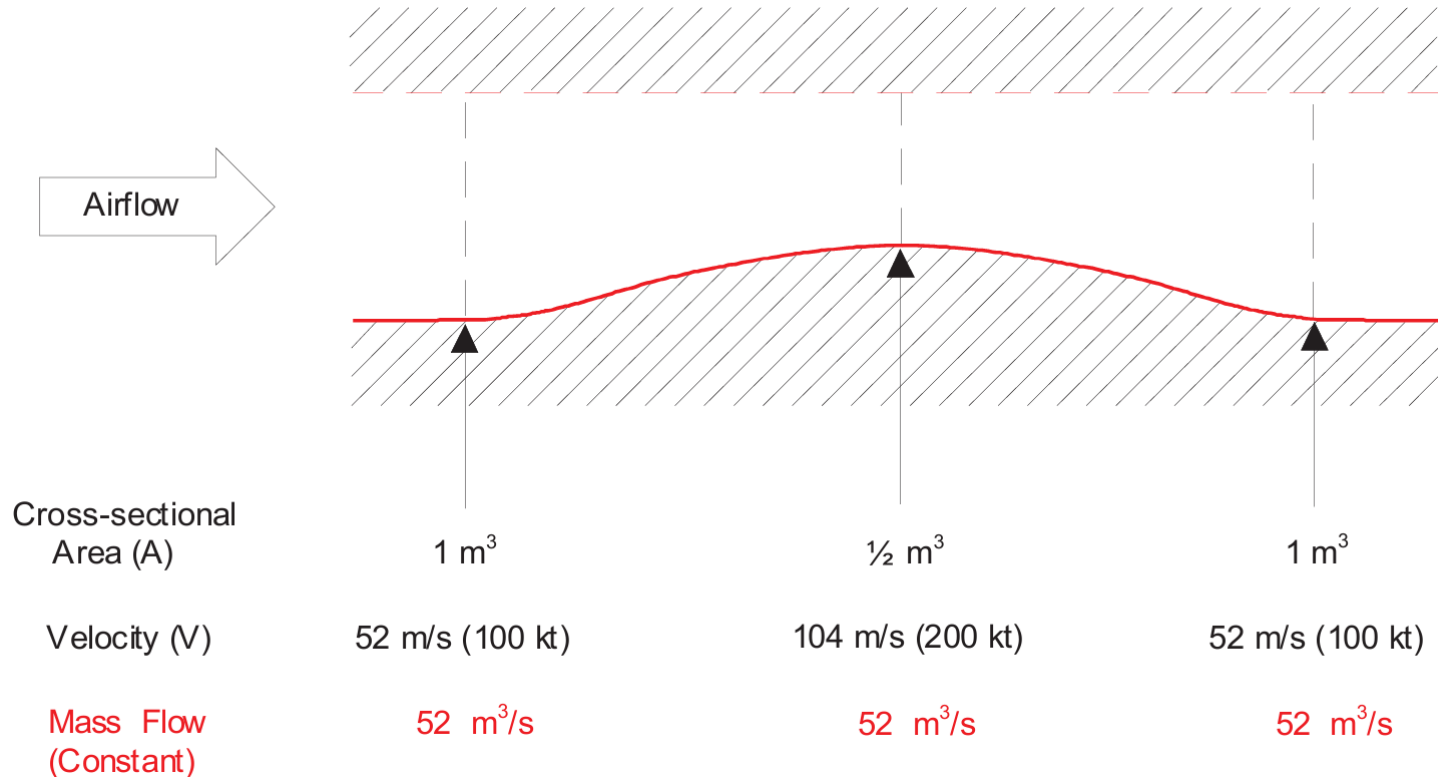
- One of the fundamental laws of the universe is ENERGY and MASS can neither be created nor destroyed, only changed from one form to another.
- The air mass flow, or mass per unit time, through the tube will be the product of the cross sectional area (A), the airflow velocity (V) and the air density (ρ). Mass flow will remain a constant value at all points along the tube. The Equation of Continuity is:

$$\mathbf{A \times V \times \rho = Constant}$$

- at low subsonic speeds ($< M 0.4$) density changes will be insignificant and can be disregarded. The equation of continuity can now be simplified to: $A \times V = \text{constant}$

$$\mathbf{Velocity (V) = \frac{Constant}{Area (A)}}$$

The Principle of Continuity



Bernoulli's Principle

Bernoulli's Theorem

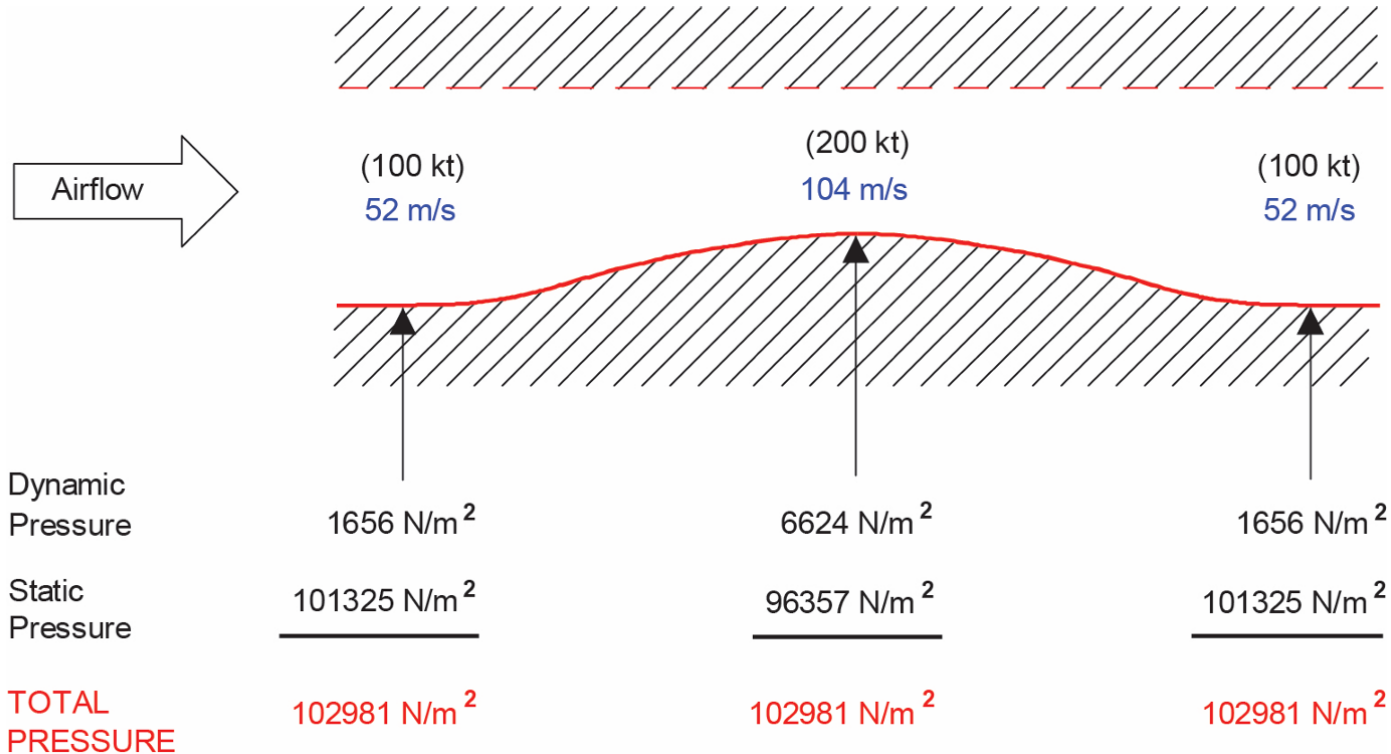
“In the steady flow of an ideal fluid the sum of the pressure energy and the kinetic energy remains constant”.

Note: An ideal fluid is both incompressible and has no viscosity.

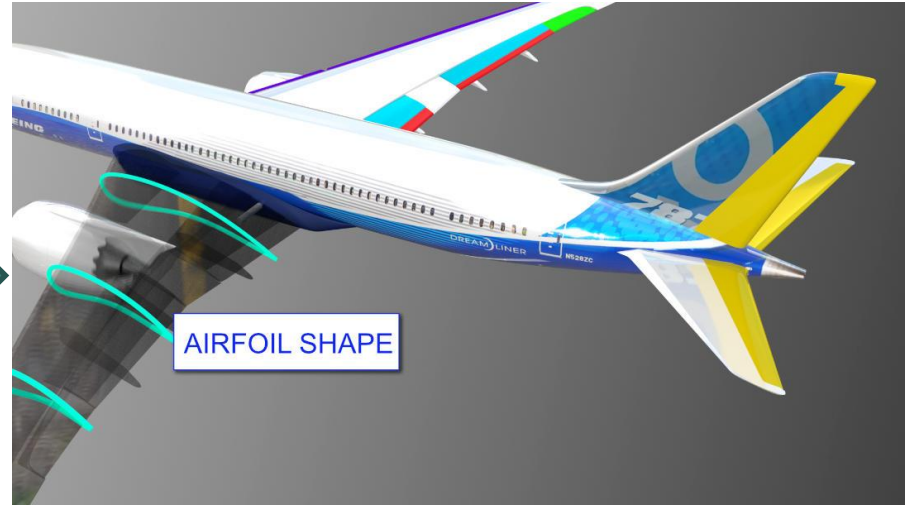
This statement can be expressed as: **Static Pressure + Dynamic Pressure = Constant**


$$p + \frac{1}{2} \rho V^2 = \text{Constant}$$

Bernoulli's Principle

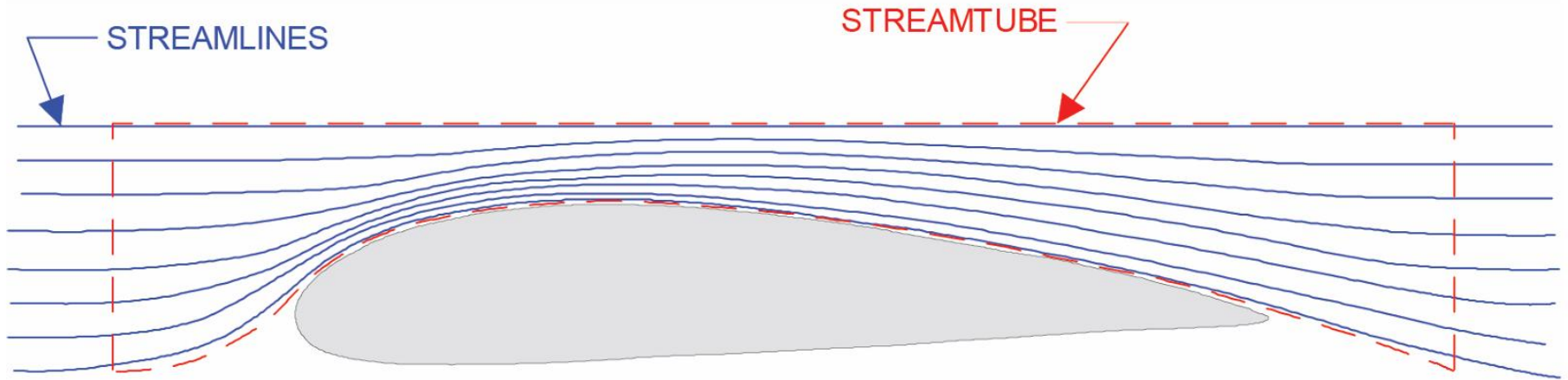


Bernoulli's Principle



The airflow velocity over the top surface of a lifting airfoil will be greater than that beneath, so the pressure differential that results will produce a force per unit area acting upwards. The larger the surface area, the bigger the force that can be generated.

Bernoulli's Principle



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SUMMARY

At flow speeds of less than about $M 0.4$, pressure changes will not affect air density.

Continuity:

- Air accelerates when the cross-sectional area of a streamline flow is reduced.
- Air decelerates when the cross-sectional area increases again.

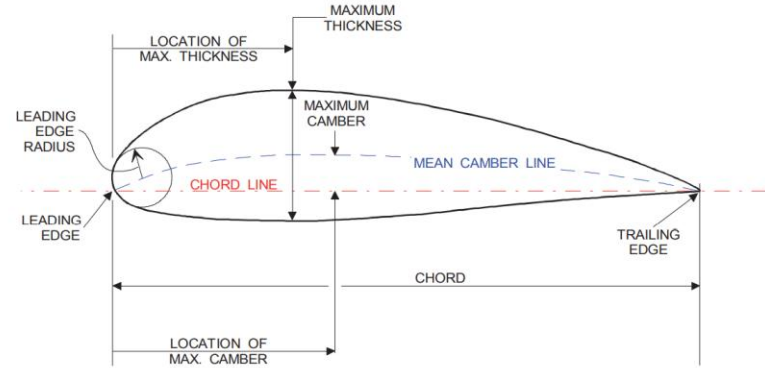
Bernoulli:

- If a streamline flow of air accelerates, its kinetic energy will increase and its static pressure will decrease.
- When air decelerates, the kinetic energy will decrease and the static pressure will increase again.

By harnessing the principle of continuity and Bernoulli's theorem an aerodynamic force can be generated.

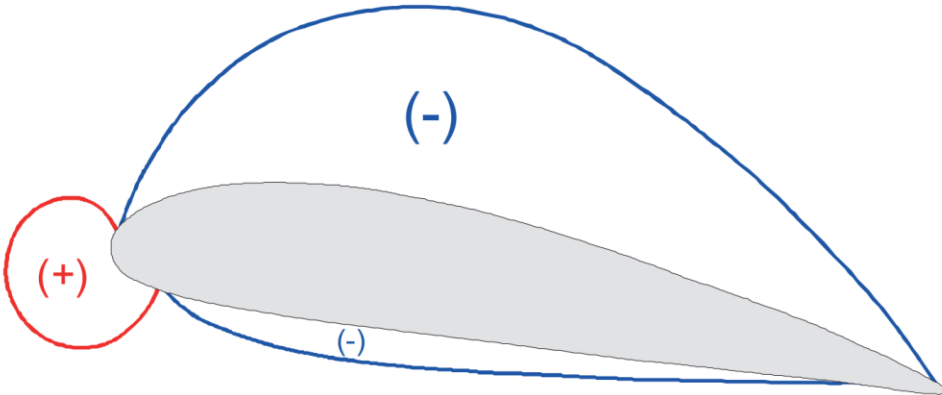
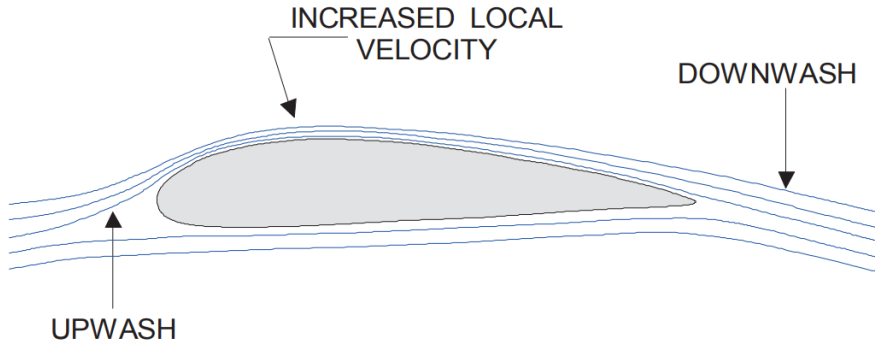
Airfoil Geometry & Definitions

- **Airfoil** : A shape capable of producing lift with relatively high efficiency.
- **Chord Line** : A straight line joining the centers of curvature of the leading and trailing edges of an airfoil.
- **Chord** : The distance between the leading and trailing edges measured along the chord line.
- **Angle of Incidence** : The angle between the wing root chord line and the longitudinal axis of the aircraft.
- **Mean Line or Camber Line** : A line joining the leading and trailing edges of an airfoil, equidistant from the upper and lower surfaces
- **Maximum Camber** : The maximum distance of the mean line from the chord line.
- **Thickness/Chord Ratio** : The maximum thickness or depth of an aerofoil section expressed as a percentage of the chord, with its location as a percentage of the chord aft of the leading edge. The thickness and thickness distribution of the aerofoil section have a great influence on its airflow characteristics.



AIRFOIL

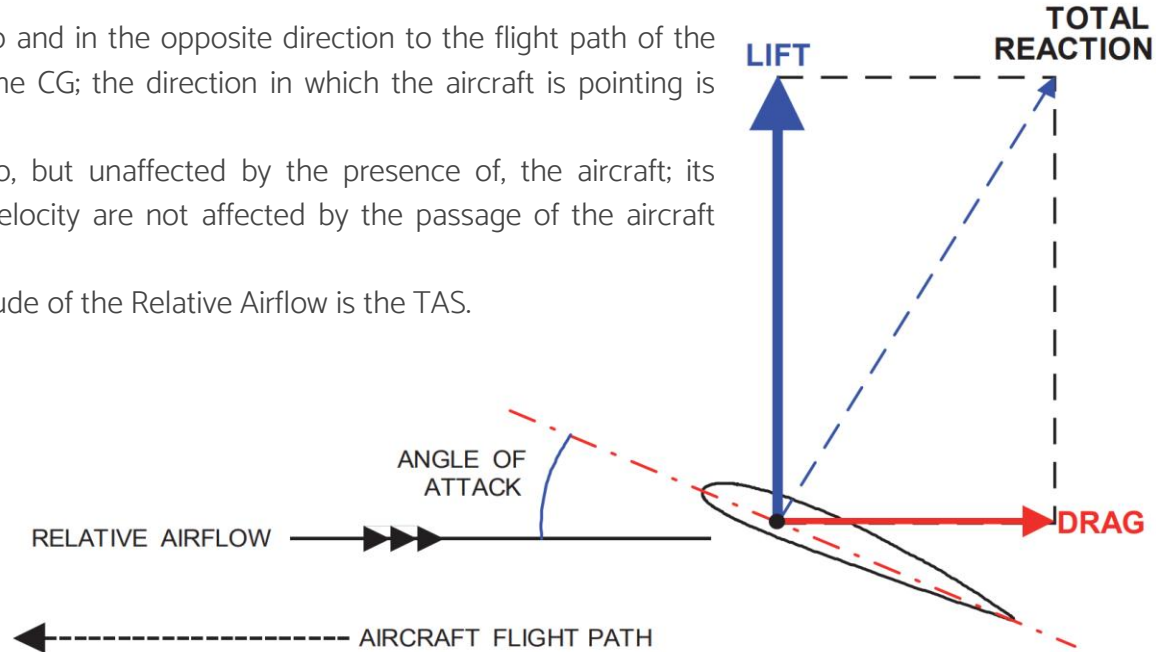
- The Figure shows an airfoil section at an angle of attack subject to a given dynamic pressure (IAS).
- “If the static pressure on one side of a body is reduced more than on the other side, a pressure differential will exist”.
- “If the dynamic pressure (IAS) is increased, the pressure differential will increase”



Relative Airflow

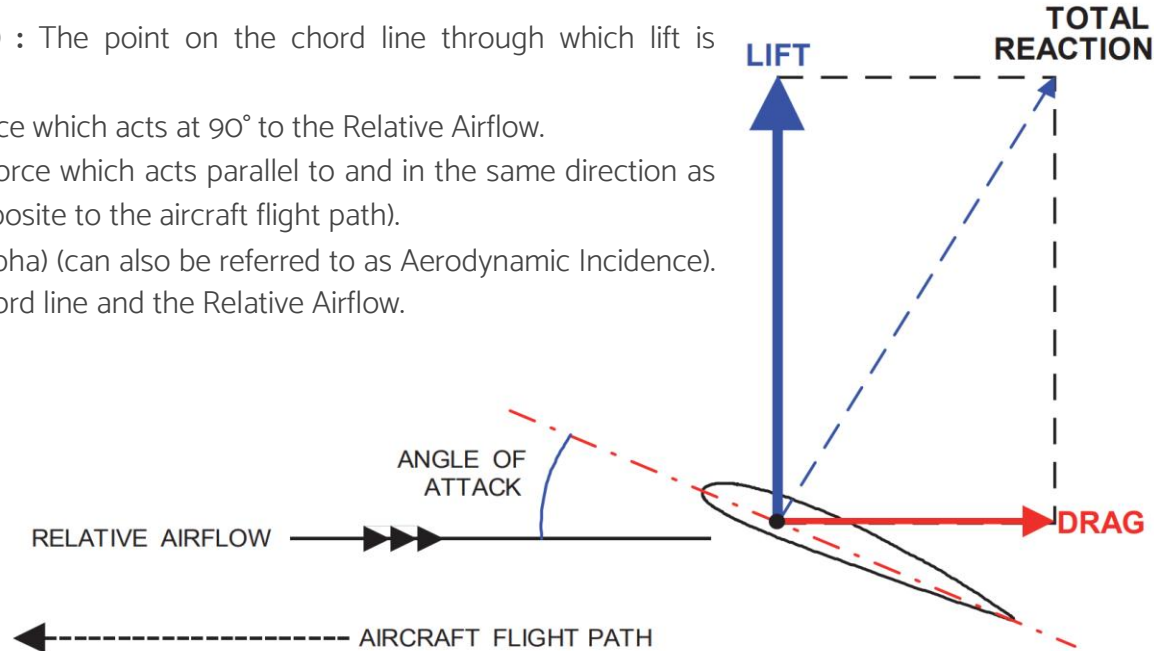
Relative Airflow (Relative Wind or Free Stream Flow) : Relative Airflow has three qualities.

- **DIRECTION** - air parallel to and in the opposite direction to the flight path of the aircraft, in fact the path of the CG; the direction in which the aircraft is pointing is irrelevant.
- **CONDITION** - air close to, but unaffected by the presence of, the aircraft; its pressure, temperature and velocity are not affected by the passage of the aircraft through it.
- **MAGNITUDE** - The magnitude of the Relative Airflow is the TAS.



Relative Airflow

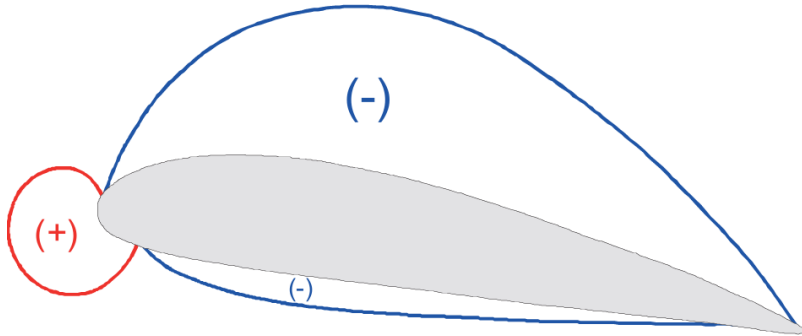
- **Total Reaction** : The resultant of all the aerodynamic forces acting on the airfoil section.
- **Centre of Pressure (CP)** : The point on the chord line through which lift is considered to act.
- **Lift** : The aerodynamic force which acts at 90° to the Relative Airflow.
- **Drag** : The aerodynamic force which acts parallel to and in the same direction as the Relative Airflow (or opposite to the aircraft flight path).
- **Angle of Attack** : (α or alpha) (can also be referred to as Aerodynamic Incidence). The angle between the chord line and the Relative Airflow.



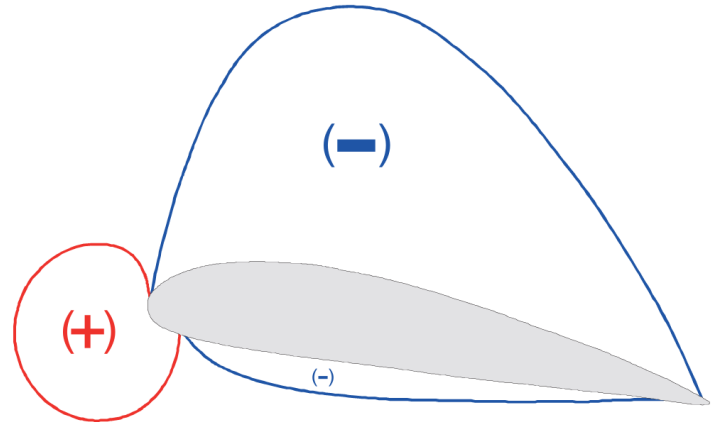
INFLUENCE OF DYNAMIC PRESSURE

- The pressure differential acting on the surface area will produce an upward acting force. “If the dynamic pressure (IAS) is increased, the upward force will increase”

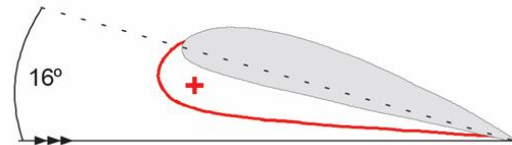
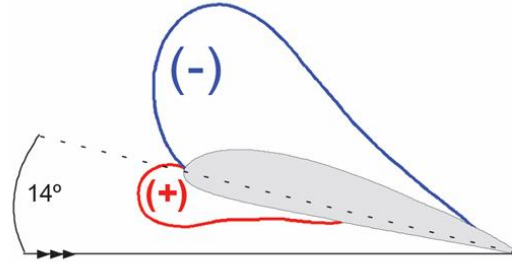
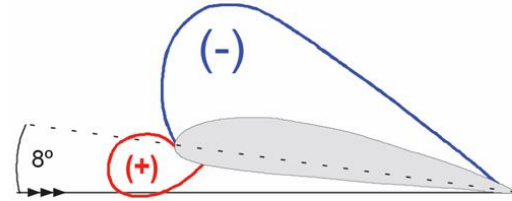
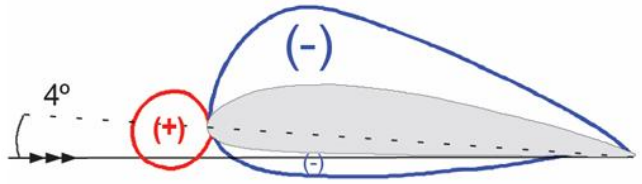
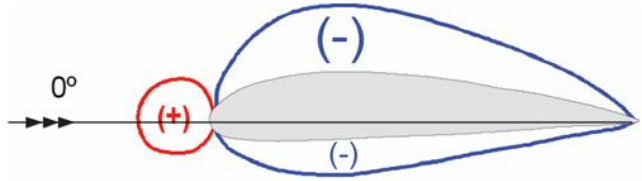
REPRESENTATIVE ANGLE OF ATTACK
AND A GIVEN DYNAMIC PRESSURE



SAME ANGLE OF ATTACK
INCREASED DYNAMIC PRESSURE



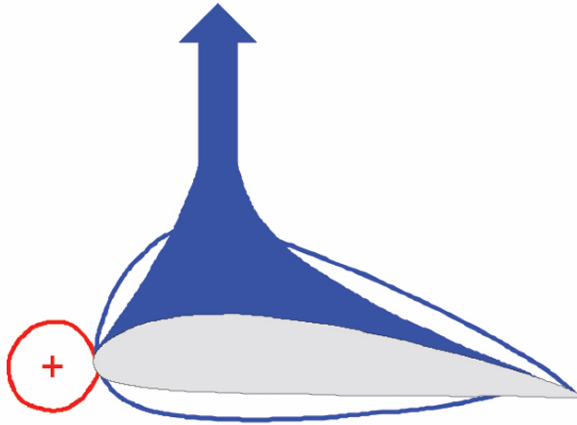
INFLUENCE OF ANGLE OF ATTACK



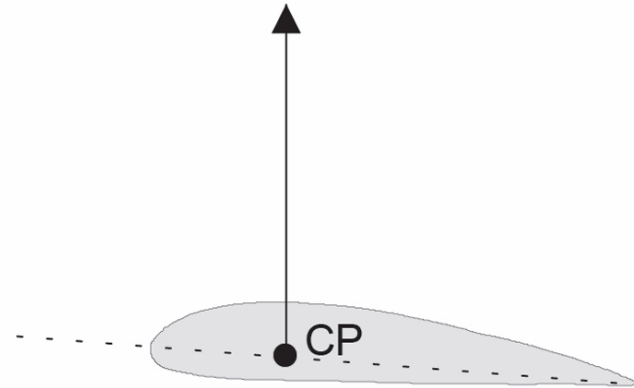
CENTER OF PRESSURE

- The whole surface of the airfoil contributes to lift, but the point along the chord where the distributed lift is effectively concentrated is termed the Centre of Pressure (Figure 4.8). The location of the CP is a function of camber and section lift coefficient, i.e. angle of attack.

AVERAGE AERODYNAMIC FORCE



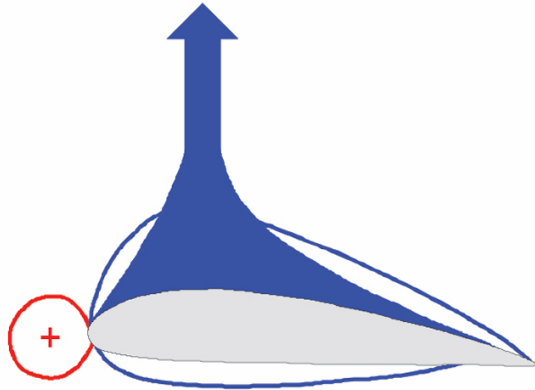
LIFT



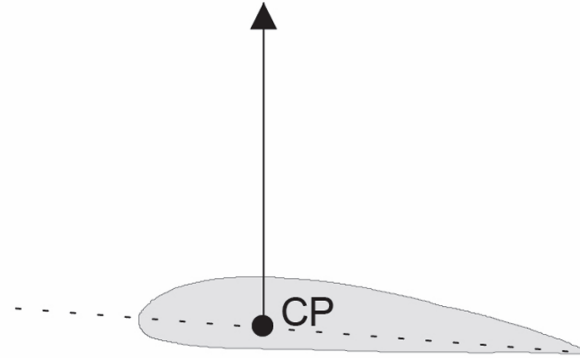
SYMMETRICAL AIRFOIL

- When at zero angle of attack, the upper and lower surface forces are equal and located at the same point.
- With an increase in angle of attack, the upper surface force increases while the lower surface force decreases.

AVERAGE AERODYNAMIC FORCE



LIFT

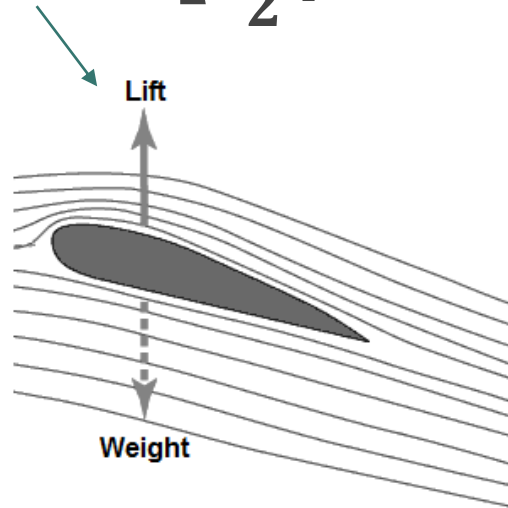


LIFT EQUATION

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$

LIFT FORCE

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$

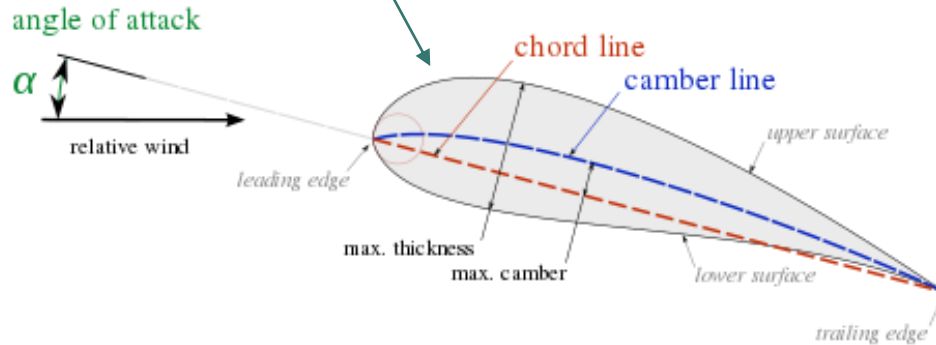


LIFT COEFFECIENT

Depends on :

- Airfoil Geometry (camber, LE, TE, etc)
- Angle of Attack

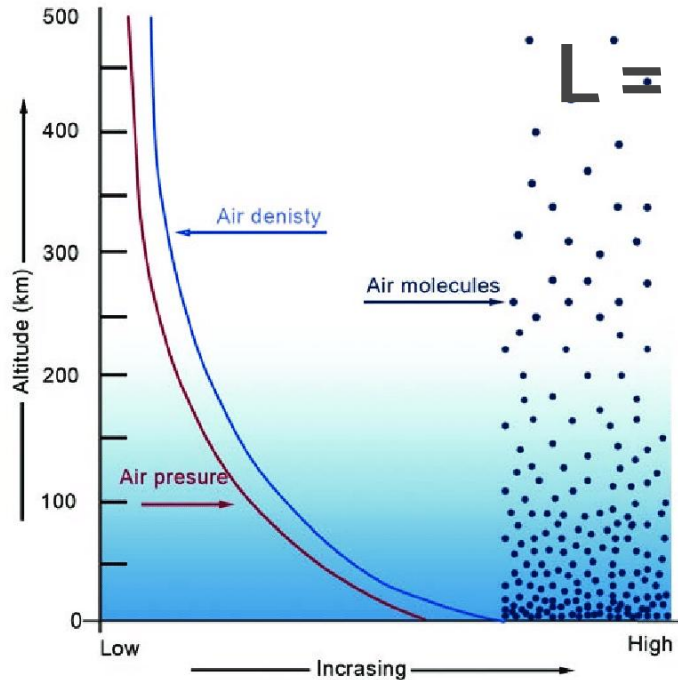
$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$



DYNAMIC PRESSURE

Depends on :

- Air Density (flight altitude)
- Airspeed



$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$

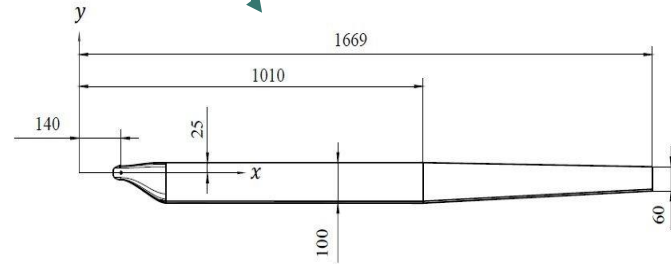
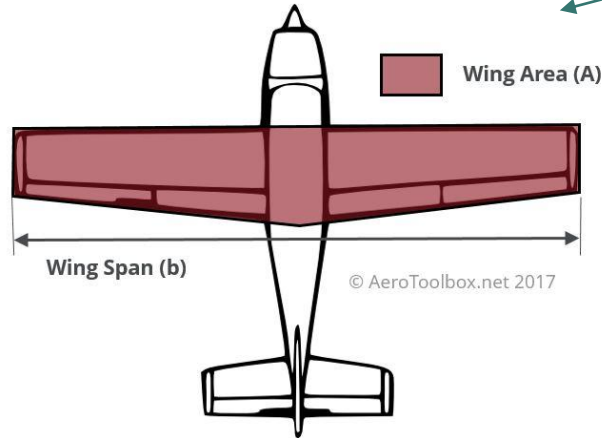


WING/BLADE AREA

Depends on :

- Wing area for fixed wings
- Rotor blade Area for helicopters

$$L = C_L \left(\frac{1}{2} \rho v^2 \right) S$$



WING/BLADE AREA

Problem 1 – Basic lift at takeoff

A small training aircraft has:

Wing area $S = 15 \text{ m}^2$

Maximum lift coefficient $C_L = 1.6$ (with flaps)

Takeoff speed $V = 25 \text{ m/s}$

Air density at the airport $\rho = 1.18 \text{ kg/m}^3$

Calculate the lift force produced by the wing at this speed.

WING/BLADE AREA

Problem 1 – Basic lift at takeoff

Given:

Wing area $S = 15 \text{ m}^2$

Maximum lift coefficient $C_L = 1.6$ (with flaps)

Takeoff speed $V = 25 \text{ m/s}$

Air density $\rho = 1.18 \text{ kg/m}^3$

Question: Calculate the lift force at this speed.

Solution:

Step 1 Write the lift equation $L = \frac{1}{2} \times \rho \times V^2 \times S \times C_L$

Step 2 Substitute the values (always write units) $L = \frac{1}{2} \times 1.18 \text{ kg/m}^3 \times (25 \text{ m/s})^2 \times 15 \text{ m}^2 \times 1.6$

Step 3 Calculate step-by-step

$$V^2 = 25 \times 25 = 625 \text{ m}^2/\text{s}^2$$

$$\frac{1}{2} \times 1.18 = 0.59$$

$$0.59 \times 625 = 368.75$$

$$368.75 \times 15 = 5\,531.25$$

$$5\,531.25 \times 1.6 = \mathbf{8\,850 \text{ N}}$$

WING/BLADE AREA

Problem 2 – Does it fly? (comparison)

The same aircraft from Problem 1 has a mass of 950 kg.

- a) Calculate its weight
- b) Compare the lift from Problem 1 to the weight. Does the wing produce enough lift to get airborne at that speed?

WING/BLADE AREA

Problem 3 – Cruise drag

An aircraft is cruising at constant speed with:

Wing area $S = 28 \text{ m}^2$

Cruise speed $V = 60 \text{ m/s}$

Drag coefficient $C_D = 0.032$

Air density $\rho = 1.225 \text{ kg/m}^3$

Calculate the thrust force the engines must provide to maintain this steady speed.

WING/BLADE AREA

Problem 4 – Lift coefficient from known conditions

A glider is in steady straight-and-level flight at 45 m/s.

Mass = 480 kg

Wing area $S = 18 \text{ m}^2$

Air density $\rho = 1.10 \text{ kg/m}^3$ (higher altitude)

Calculate the required lift coefficient C_L at this speed.

WING/BLADE AREA

Problem 4 – Lift coefficient from known conditions

Given

Glider mass = 480 kg

Wing area $S = 18 \text{ m}^2$

Air density $\rho = 1.10 \text{ kg/m}^3$

Airspeed $V = 45 \text{ m/s}$

In steady level flight $\rightarrow$ Lift = Weight

Question: Calculate the required lift coefficient C_L .

Solution:

Step 1 Calculate the weight (this is the lift we need) $\text{Weight} = m \times g = 480 \text{ kg} \times 9.81 \text{ m/s}^2 = \mathbf{4\,708.8 \text{ N}}$ $\rightarrow$ So Lift $L = 4\,708.8 \text{ N}$

Step 2 Rearrange the lift equation to solve for C_L $L = \frac{1}{2} \times \rho \times V^2 \times S \times C_L \rightarrow C_L = L / (\frac{1}{2} \times \rho \times V^2 \times S)$ or more cleanly:

$$C_L = (2 \times L) / (\rho \times V^2 \times S)$$

Step 3 Substitute values $C_L = (2 \times 4\,708.8) / (1.10 \times 45^2 \times 18)$

Calculate inside:

$$45^2 = 2\,025$$

$$1.10 \times 2\,025 = 2\,227.5$$

$$2\,227.5 \times 18 = 40\,095$$

$$2 \times 4\,708.8 = 9\,417.6 \rightarrow C_L = 9\,417.6 / 40\,095 \approx \mathbf{0.235}$$

WING/BLADE AREA

Problem 5 – Speed change with altitude (density effect)

The same glider from Problem 4 now climbs to a higher altitude where air density drops to $\rho = 0.85 \text{ kg/m}^3$. The pilot wants to keep producing the same lift force (to maintain level flight at same mass).

Assume C_L stays the same (same angle of attack / configuration).

Calculate the new airspeed needed to generate the same lift.

WING/BLADE AREA

Problem 6 – Required speed for given lift

A light sport aircraft needs to produce 8,500 N of lift to stay level.

Wing area $S = 12 \text{ m}^2$

$C_L = 0.95$ (clean configuration)

$\rho = 1.225 \text{ kg/m}^3$ (sea level)

Calculate the minimum steady flight speed (V) needed to generate this lift.